

Assignment 11

Shahriar Hasan

Id: 2010776115

July 22, 2025

1 Comparing three face verifiers which are trained using Binary Cross-Entropy loss, contrastive loss and triplet loss, receptively.

1. Binary Cross-Entropy Loss
2. Contrastive Loss
3. Triplet Loss

Each model is trained on the LFW-Funneled dataset and evaluated based on accuracy and AUC (Area Under the ROC Curve).

1.1 Dataset

1.1.1 Source

The experiments use the LFW-Funneled dataset from Kaggle ([angelproject/lfw-dataset](https://www.kaggle.com/angelproject/lfw-dataset)). The dataset includes:

- `lfw-funneled.tgz`: aligned face images
- `pairsDevTrain.txt`: training image pairs
- `pairsDevTest.txt`: testing image pairs

1.1.2 Preprocessing

- Image size: 105×105
- Normalization: pixel values scaled to $[0, 1]$
- Data split: 80% training, 20% validation (from `pairsDevTrain.txt`)
- Testing on `pairsDevTest.txt`

1.2 Model Architecture

All models share a common base network composed of:

- 4 convolutional layers
- 3 max-pooling layers
- 1 fully connected layer (4096 units, sigmoid activation)

1.2.1 Binary Cross-Entropy (BCE) Siamese Network

- Computes absolute difference between feature vectors
- Trained using Binary Cross-Entropy Loss
- Outputs a probability of whether two images belong to the same identity

1.2.2 Contrastive Network

- Uses Euclidean distance
- Trained using Contrastive Loss
- Encourages similar pairs to have small distances and dissimilar pairs to have larger distances

1.2.3 Triplet Network

- Trained using triplets: anchor, positive, and negative
- Uses Triplet Loss
- Optimizes embeddings so that positive examples are closer to the anchor than negative ones

1.3 Evaluation Metrics

- **Accuracy:** Percentage of correctly classified pairs
- **AUC:** Area Under the ROC Curve, indicates ranking/separability

1.4 Results

Model	Accuracy (%)	AUC
Binary Cross-Entropy (BCE)	59.50	0.7090
Contrastive Loss	61.30	0.7552
Triplet Loss	51.50	0.7740

Table 1: Performance Comparison of Face Verification Models

1.5 ROC Curve Visualization

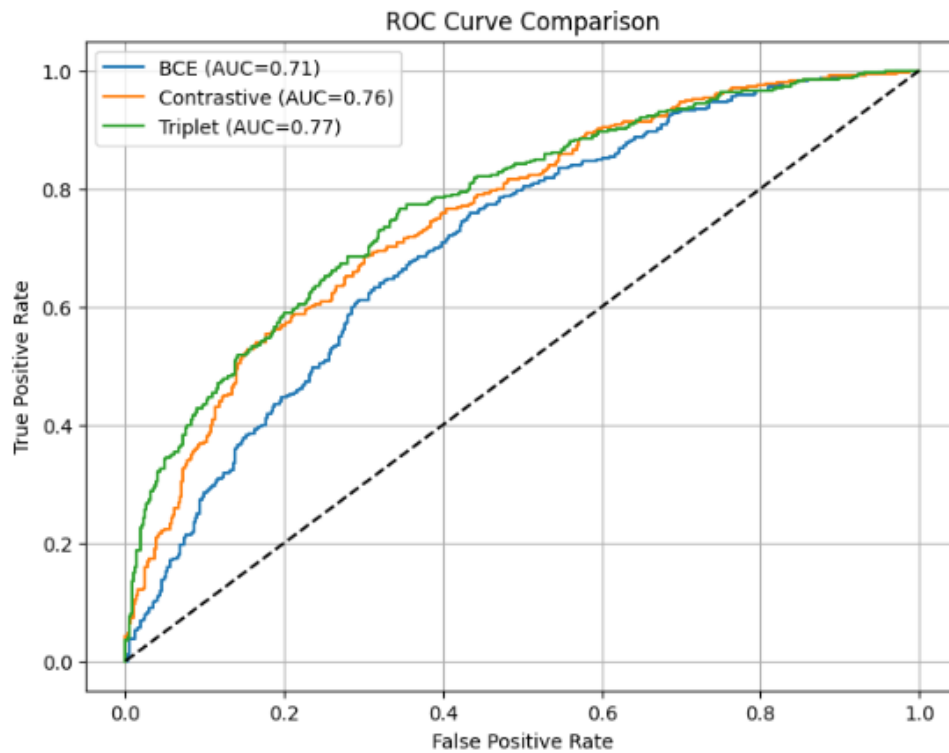


Figure 1: ROC Curve Comparison for BCE, Contrastive, and Triplet Models

1.6 Analysis

1.6.1 Binary Cross-Entropy (BCE)

- Moderate accuracy and AUC
- Simple to implement but weaker pairwise separability

1.6.2 Contrastive Loss

- Achieved the highest accuracy
- Well-suited for pairwise similarity comparison

1.6.3 Triplet Loss

- Best AUC score, indicating stronger ranking capability
- Accuracy is limited, possibly due to the lack of hard triplet mining

2 Comparing effect of Binary Cross-Entropy (BCE) loss and Mean Squared Error (MSE) as a reconstruction loss during training of a Variational Autoencoder (VAE).

2.1 Methodology

Two VAE models with identical architectures were implemented, differing only in the reconstruction loss function used:

- **BCE VAE:** Reconstruction loss calculated using Binary Cross-Entropy.
- **MSE VAE:** Reconstruction loss calculated using Mean Squared Error.

Both models comprise an encoder producing latent mean and log-variance vectors, a reparameterization step to sample latent vectors, and a decoder reconstructing images. The total loss combines the reconstruction loss and a Kullback-Leibler (KL) divergence regularizer, encouraging the latent distribution to approximate a standard normal.

The models were trained on the MNIST dataset for 10 epochs using the Adam optimizer with a batch size of 128.

2.2 Results

2.2.1 Training Loss Curves

- The **BCE VAE** showed faster convergence in total and reconstruction losses early in training.
- The **MSE VAE** exhibited smoother loss curves but converged more slowly.
- The KL divergence loss was similar for both models, indicating comparable regularization.

2.3 Reconstruction Quality

Qualitative inspection of reconstructed images reveals:

- **BCE VAE** reconstructions are sharper and better preserve digit edges and details.
- **MSE VAE** reconstructions tend to be blurrier, likely due to squared error penalization of pixel-wise differences.

2.4 Figures

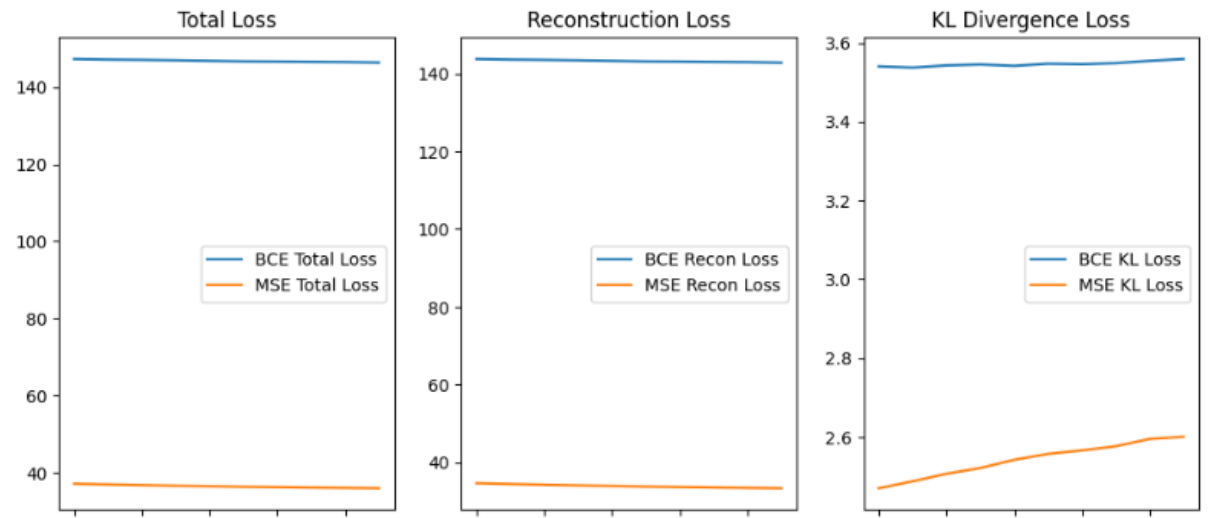


Figure 2: Training loss curves for BCE and MSE VAEs: total loss, reconstruction loss, and KL divergence.

Code Availability

The complete implementation is available on GitHub:

- **Github Link**

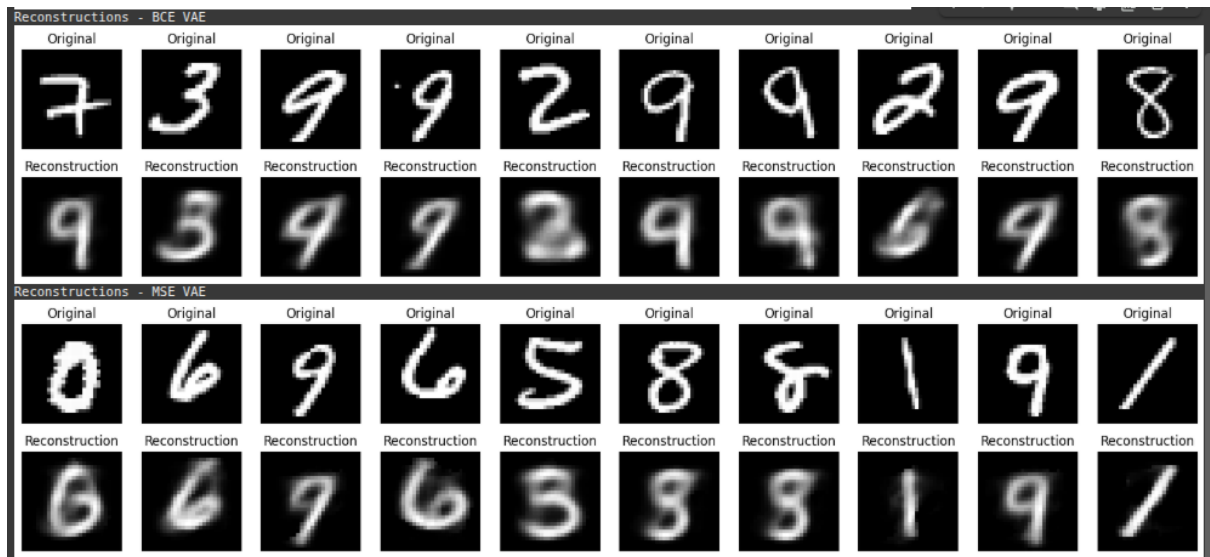


Figure 3: Original and reconstructed MNIST digits for BCE and MSE VAEs. Top row: original images. Middle row: BCE VAE reconstructions. Bottom row: MSE VAE reconstructions.